Equation

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$$
\begin{aligned}
\text{PercentOrganicSeed}_i &\sim N\left(\mu, \sigma^2\right) \\
\mu &= \alpha_{j[i],k[i],l[i]} + \beta_1(\text{Values}) + \beta_2(\text{Acreage}) + \beta_3(\text{CertifierRequest}) \\
\alpha_j &\sim N\left(\gamma_0^\alpha + \gamma_1^\alpha(\text{Funding}) + \gamma_2^\alpha(\text{Growers:Suppliers}) + \gamma_3^\alpha(\text{Revenue}), \sigma^2_{\alpha_j}\right), \text{ for Region j} = 1,\ldots,\text{J} \\
\alpha_k &\sim N\left(\mu_{\alpha_k}, \sigma^2_{\alpha_k}\right), \text{ for Year k} = 1,\ldots,\text{K} \\
\alpha_l &\sim N\left(\mu_{\alpha_l}, \sigma^2_{\alpha_l}\right), \text{ for Crop l} = 1,\ldots,\text{L}
\end{aligned}
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$$
\begin{aligned}
\text{IncreasedOrganicSeed}_i &\sim \text{Binomial}(n = 1, \text{prob}_{\text{IncreasedOrganicSeed}=1} = \widehat{P}) \\
\log\left[\frac{\hat{P}}{1-\hat{P}}\right] &= \alpha_{j[i],k[i],l[i]} + \beta_1(\text{Values}) + \beta_2(\text{Acreage}) + \beta_3(\text{CertifierRequest}) \\
\alpha_j &\sim N\left(\gamma_0^\alpha + \gamma_1^\alpha(\text{Funding}) + \gamma_2^\alpha(\text{Growers:Suppliers}) + \gamma_3^\alpha(\text{Revenue}), \sigma^2_{\alpha_j}\right), \text{ for Region j} = 1,\ldots,\text{J} \\
\alpha_k &\sim N\left(\mu_{\alpha_k}, \sigma^2_{\alpha_k}\right), \text{ for Year k} = 1,\ldots,\text{K} \\
\alpha_l &\sim N\left(\mu_{\alpha_l}, \sigma^2_{\alpha_l}\right), \text{ for Crop l} = 1,\ldots,\text{L}
\end{aligned}
$$